

GATEWAYS 2026
Round 1 — Ideation & Architecture Document

ResearchOps

The Autonomous Research Agent that Investigates, Argues, Verifies and Decides

1. Team details

Field	Details
Team Name	CodeBlooded
Problem Statement Chosen	ResearchOps: The Autonomous Research Agent
Team Members (Name, Reg No., Role)	Samrat S K (1BM24CS253): AI/Agent Architecture and Backend
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2. Problem understanding

Businesses have more information available than ever, but turning it into **reliable research** is still a manual, fragmented job. **Analysts, founders, product managers, consultants and strategy teams** investigate markets, competitors, technologies, customers and regulations by searching company websites, news, reports, government sources and forums. They must decide what is relevant, extract facts, **compare sources**, check whether information is **current**, spot **contradictions**, and organise everything into a report. Knowledge workers spend roughly **1.8 hours a day, about 19% of their time, searching for and gathering information** [1].

The hard part is not finding information. It is **deciding how much to trust it**. When the question is broad or ambiguous, the researcher decides alone what to investigate and when to stop. Important facts get missed, **outdated or conflicting claims go unnoticed**, ten articles repeating one press release look like strong agreement, and the conclusion ends up depending on whichever sources were found first.

Why it matters: real-world cases

Deciding without understanding the market. Analyses of startup post-mortems consistently rank **"no market need"** among the top reasons companies fail, cited in **42%** of failures in CB Insights' original study [2]. **Quibi** raised about **US\$1.75 billion** and shut down roughly six months after launch in 2020, after betting that people wanted premium short-form video on their phones [3]. In both cases the evidence against the idea existed; it was not gathered, weighed and challenged before money was committed.

Believing the most repeated claim. Amazon's **"Just Walk Out"** cashier-less checkout was marketed as computer-vision automation. In 2024 reports stated that it relied on **more than 1,000 workers in India** reviewing footage, with about **700 of every 1,000 transactions** needing human review against an internal target of 50 [4][5]. Amazon disputed this characterisation as misleading [5] and replaced the system in its US Fresh stores with **Dash Carts** [4]. A retailer researching automated checkout would have found mostly vendor claims repeated across many articles, and a smaller number of independent reports saying something very different. That is exactly the kind of **disagreement between interested and independent sources** that manual research, and today's AI tools, tend to flatten.

Trusting AI-written research without verification. In 2025 **Deloitte** agreed to partially refund the Australian government for an **A\$440,000** report for the Department of Employment and Workplace Relations after it was found to contain **AI-generated fabricated references and quotes**, including a paper that does not exist [6]. Benchmarks of commercial deep-research agents show this is systemic: the best systems reach only about **65% citation quality** and **68% factual**

accuracy [7], 3–13% of cited URLs are fabricated, and true statements are often attached to the **wrong source [8][9]**.

The real problem

Research is the foundation of decisions about **investment, product launches, expansion, pricing and partnerships**. The challenge is not simply finding information but building a **reliable chain** from an open-ended question to focused research tasks, relevant evidence, cross-source verification, and a conclusion that can be **traced back to its sources**, with an honest statement of **how sure** it is.

Summary:

Decision-makers need research they can **act on and verify**. ResearchOps turns an open question into a **decision with a stated probability**, where every sentence is backed by a **verbatim quote**, every disagreement is visible, and the few facts that could change the decision are named.

3. Proposed solution — overview

ResearchOps is an **autonomous research agent** that runs research the way a careful team would, and then puts its own findings on trial. It does not just search for an answer. It first makes sure it understands **what you are trying to decide**, breaks the question into focused research tasks, collects evidence from many sources, **challenges** that evidence, and only then gives a conclusion together with the proof behind it.

How it works, in plain terms

- **It listens first.** ResearchOps asks open questions about your goal in your own words, with no fixed forms, and **does not start until it understands the intent**. From this it forms a few clear **hypotheses**, such as "there is unmet demand" or "current players leave a price gap", each with a starting likelihood.
- **It investigates in parallel.** The question becomes a **tree of research tasks**, searched **broadly first and then in depth**, with many tasks running at the same time.
- **It keeps an evidence ledger.** Every finding is stored as a short claim with the **exact sentence from the source**, the link, the source type and the date. Copies of the same story are **recognised and counted once**.
- **It chases what matters most.** When evidence is missing, it researches the gap that could **change the conclusion the most**, and it asks you a question only when your answer would change the outcome.
- **It holds a trial.** An **advocate** argues for, a **challenger** argues against, **expert witnesses** speak on market, competitors, regulation and customers, and a **pre-mortem witness** imagines the plan failed a year later and explains why. They may only argue with evidence from the ledger.
- **It verifies.** A separate **citation verifier** checks every quoted claim and strikes out anything its source does not actually say.
- **It decides.** The report opens with a **verdict and a probability**, for example "64% favourable", then gives **what must be done, what must not be done**, where sources **disagree**, what could **not be verified**, and the **two or three facts that would flip the decision**. You can adjust the assumptions it inferred from your goals with **what-if sliders** and watch the verdict recalculate instantly.

How this solves the cases above

Real-world failure	What ResearchOps does differently
No market need (Quibi, CB Insights)	Demand is an explicit hypothesis with a probability. The challenger and pre-mortem witness are required to look for evidence against it, so a missing market is surfaced before money is spent.

Real-world failure	What ResearchOps does differently
Repeated vendor claims (Just Walk Out)	Source independence detection folds repeated marketing into one piece of evidence. Independent reports are weighted separately, and the contradiction is shown side by side with dates and source types.
Fabricated citations (Deloitte report)	Nothing enters the ledger without a verbatim quote from a fetched page , so invented sources cannot appear. The verifier then checks that the quote supports the claim.
Outdated facts	Each claim is dated and loses weight over time through freshness decay ; stale evidence is labelled.
Unknown unknowns	Gaps are recorded under " What we could not verify " instead of being guessed, and the verdict's uncertainty range stays wider.

In one line

Other research tools write an essay with links; **ResearchOps produces a decision with odds and shows the evidence behind every word.**

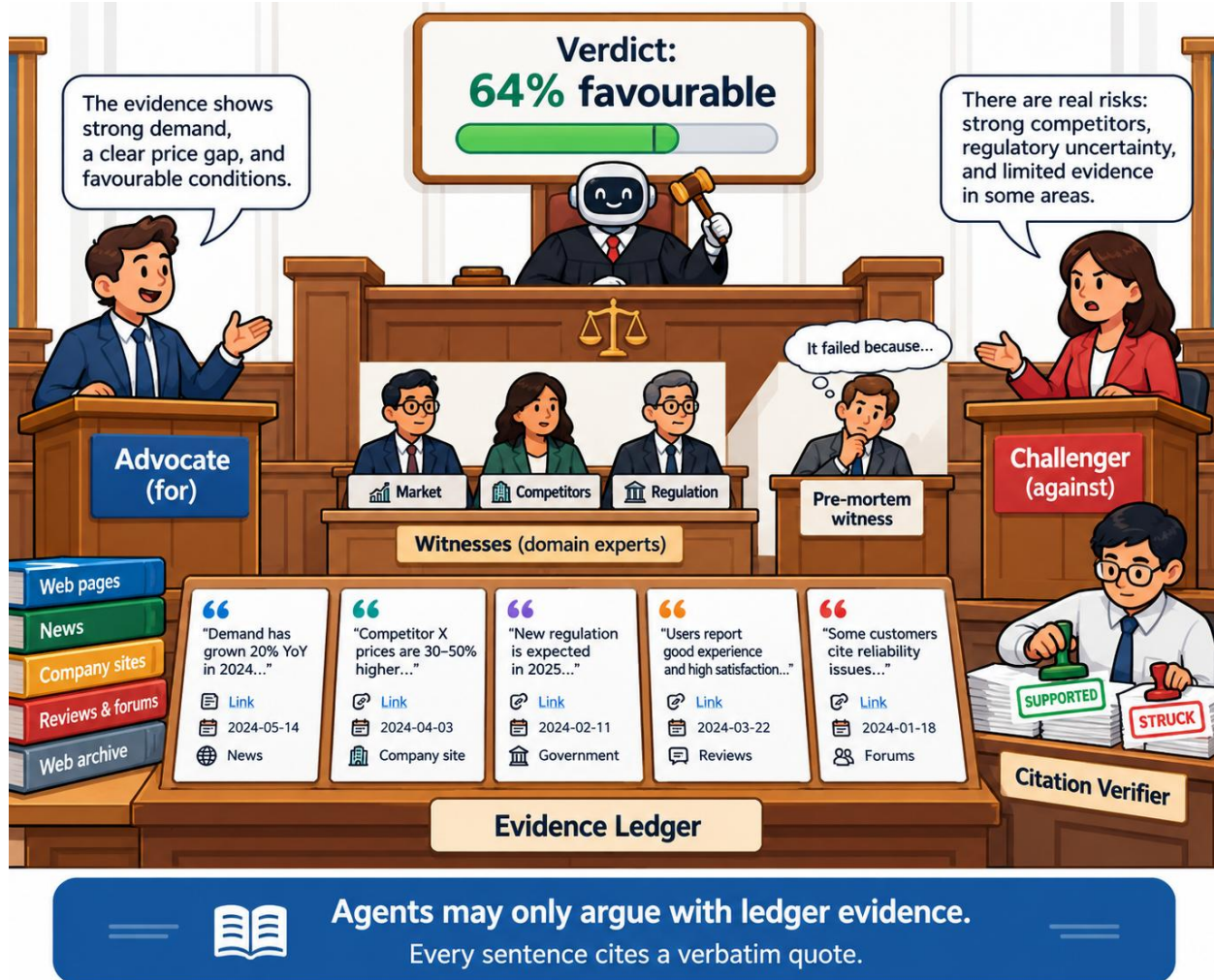


Figure 3.1: The ResearchOps Trial

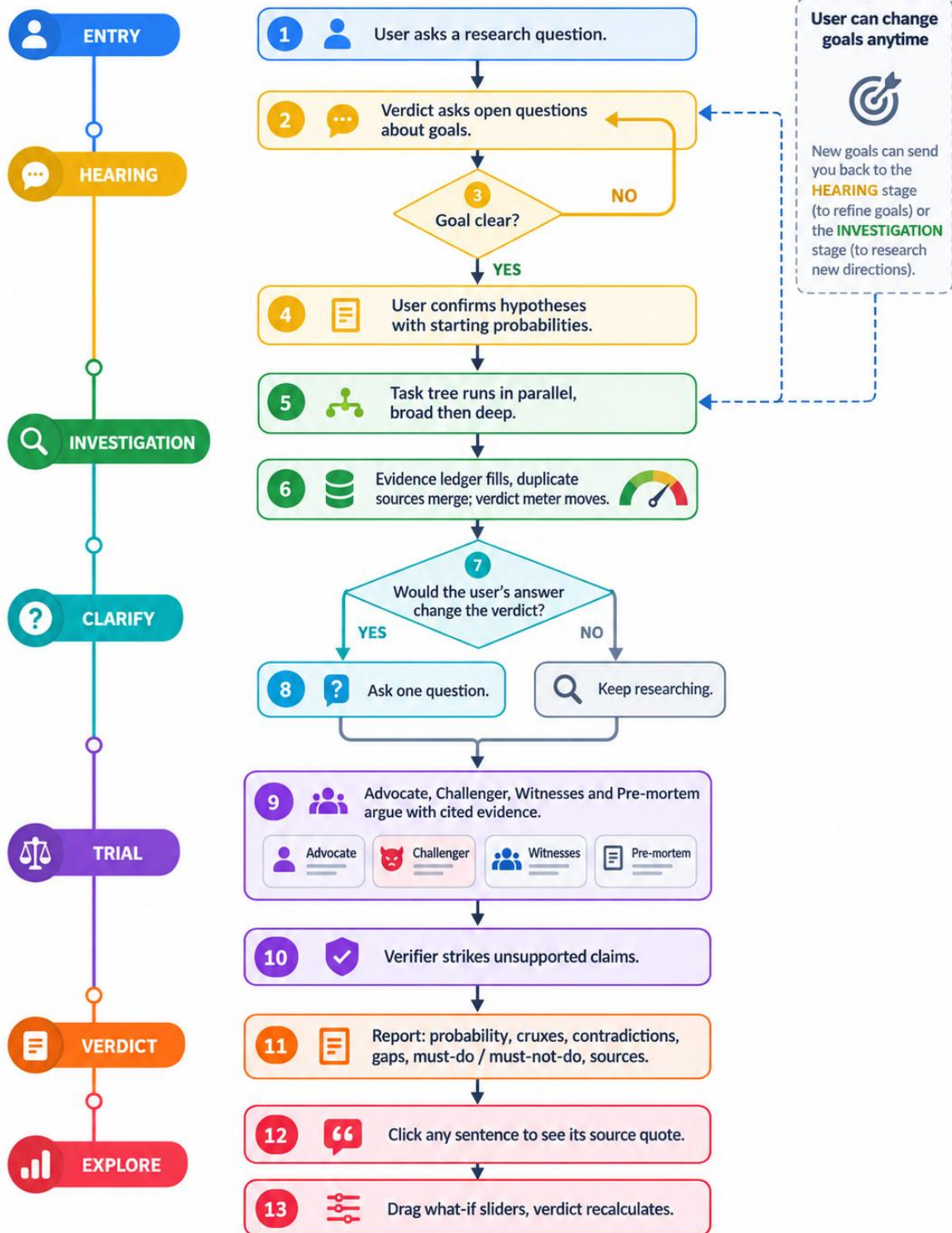


Figure 3.2: User Flow Diagram

4. System architecture

ResearchOps is organised in five layers. The user talks only to the **research workspace**. The **Research Director** owns each run and decides what happens next. **Research agents** gather evidence, **reasoning agents** argue about and score it, and the **evidence store** is the single source of truth that every other part reads from.

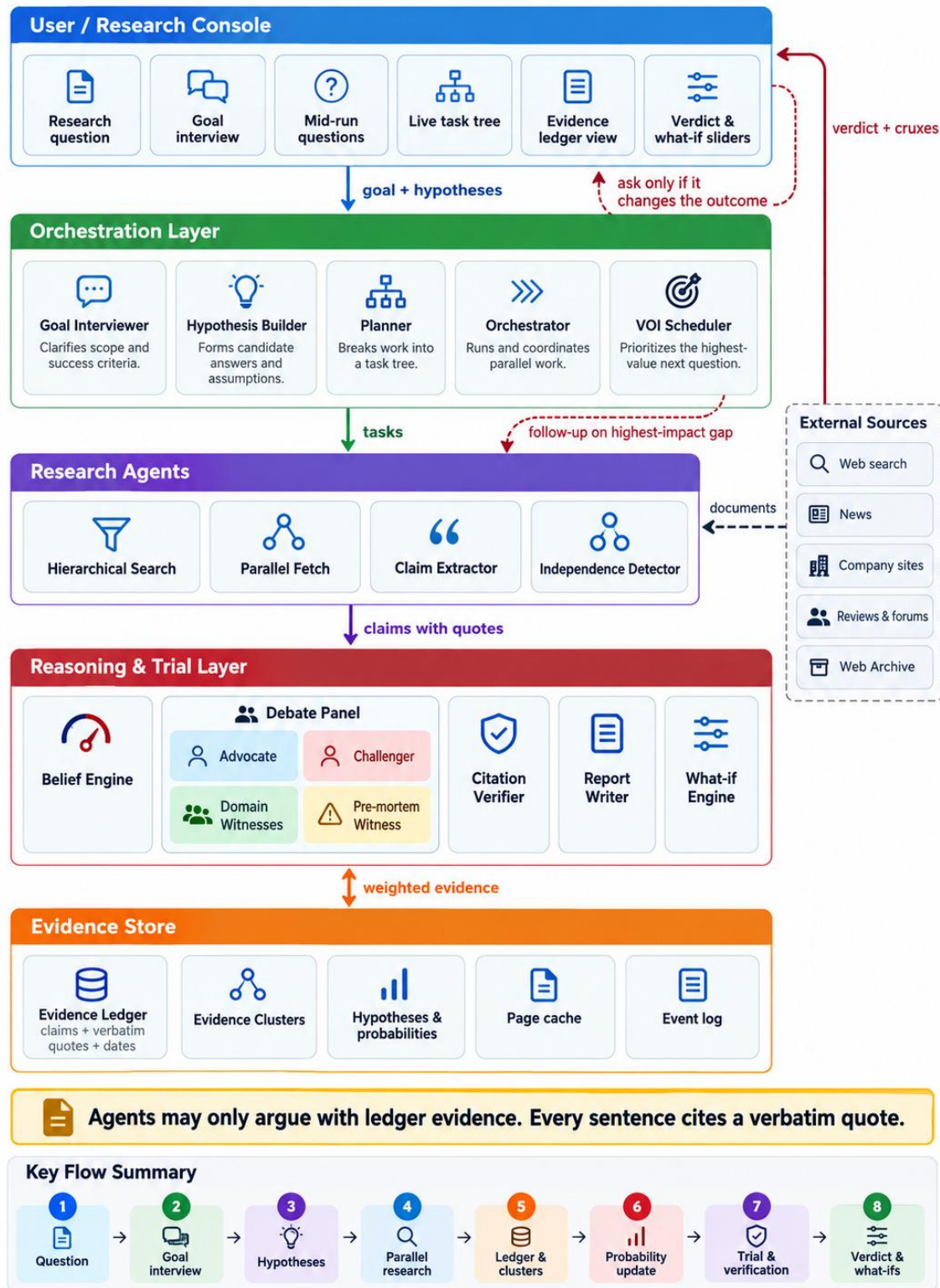


Figure 4.1: Architecture Diagram

Components and responsibilities

Component	Responsibility	Input -> output
Goal Interviewer	Open-ended conversation that extracts intent and scope; blocks the run until intent is confirmed.	User -> confirmed goal
Hypothesis Builder	Turns the goal into testable hypotheses with priors and the assumptions behind them.	Goal -> hypotheses
Research Director	Decomposes the question into a task tree , runs tasks in parallel , keeps research within scope, and re-plans when the goal changes.	Hypotheses -> tasks
Search Agents	Hierarchical search (broad to deep) across web, news, company pages, reviews and forums, with caching and provider fallback.	Tasks -> documents
Claim Extractor	Pulls atomic claims with verbatim quotes , entity, attribute, date and source type.	Documents -> ledger
Independence Detector	Groups near-duplicate claims and shared origins into one evidence cluster .	Ledger -> clusters
Belief Engine	Updates each hypothesis probability from weighted, independent evidence .	Clusters -> probabilities
VOI Scheduler	Ranks gaps by how much they could move the verdict (value of information); spawns follow-up tasks or asks the user.	Probabilities -> tasks / questions
Debate Panel	Advocate, challenger, domain witnesses and pre-mortem witness , limited to ledger citations.	Ledger -> arguments
Citation Verifier	Independent check that each quote supports its claim; strikes unsupported claims .	Arguments -> rulings
Report Writer	Builds the decision report; every sentence cites ledger IDs ; separates facts, derived insights and open uncertainties.	All -> report
What-if Engine	Recomputes the verdict when the user adjusts an inferred assumption .	Assumptions -> verdict

Information flow

1. The question and the user's answers go to the **Goal Interviewer**, which produces a confirmed goal and scope.
2. The **Hypothesis Builder** forms hypotheses, priors and assumptions; the user confirms or edits them.
3. The **Research Director** creates a task tree and runs **search agents in parallel**.

4. Documents become **ledger claims with quotes**; the **Independence Detector** clusters them.
5. The **Belief Engine** updates probabilities; the **VOI Scheduler** picks the next gap or asks the user.
6. When the verdict stabilises, the **Debate Panel** argues and the **Citation Verifier** rules on each claim.
7. The **Report Writer** produces the verdict; the **What-if Engine** keeps it interactive.

Every step emits a **live event** with the data it produced, so the workspace shows exactly how the conclusion was reached.

How the probability is calculated

The language model **reads and classifies** evidence; **transparent code computes** the probability, so the number is not invented by the model. There are **no predefined fields** such as budget or risk appetite: hypotheses and assumptions come from the user's own goal for each run. Each evidence cluster shifts a hypothesis according to four factors:

- **Direction and strength:** whether it supports or weakens the hypothesis, and how directly.
- **Source reliability:** official data and independent reporting count more than vendor marketing or anonymous posts.
- **Freshness:** each kind of fact has a **half-life**; a price loses weight within months, a regulation lasts longer.
- **Independence:** a cluster of copies counts **once**, so repetition cannot inflate confidence.

Updates are combined in **log-odds** form, so each piece of evidence has a visible, reversible effect. The verdict shows a probability and a **range** that narrows as independent evidence accumulates. After the verdict, the engine tests which clusters or assumptions would **flip the decision**; these become the **cruxes**.

Evidence-to-Verdict Data Flow

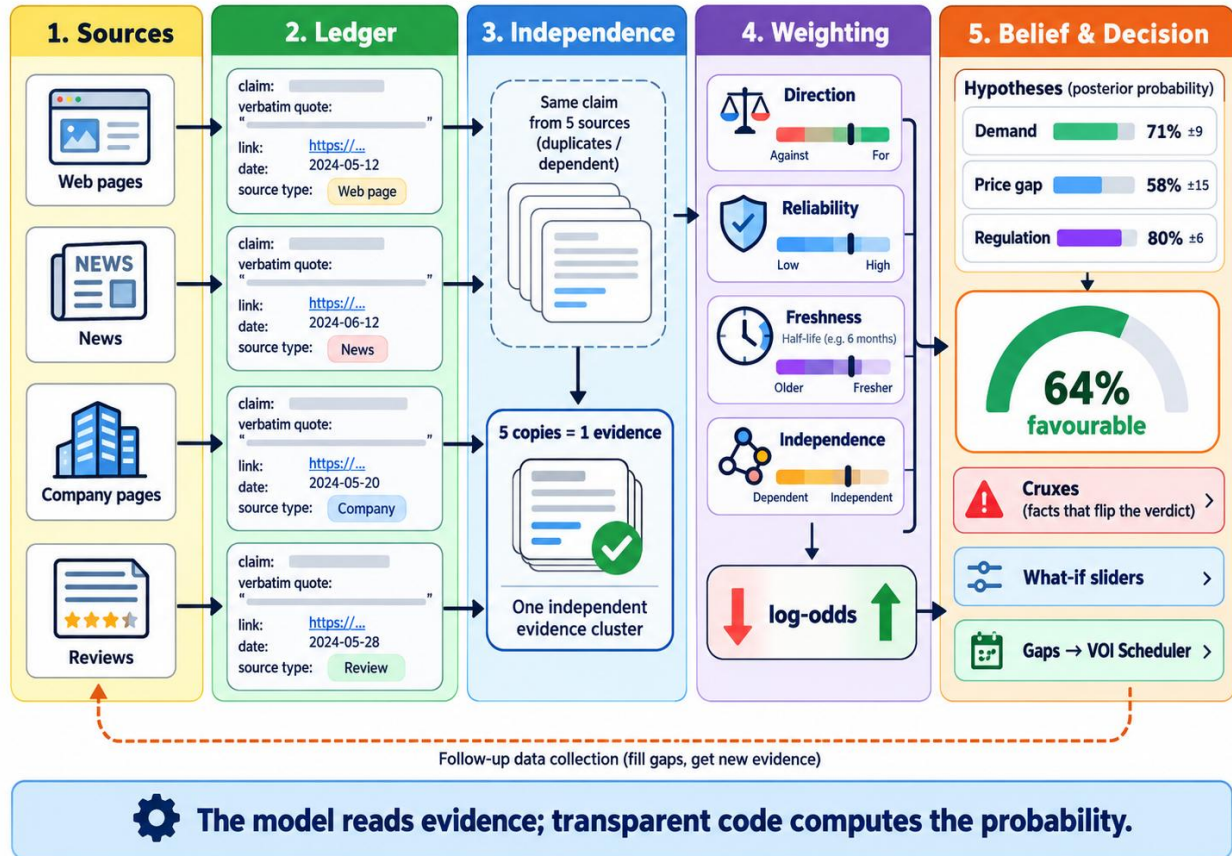


Figure 4.2: Evidence-to-Verdict Data Flow

5. Tech stack

Layer	Choice
Languages	Python, TypeScript/JavaScript, SQL
Frontend	React with Vite : research workspace, live task tree and agent status, evidence ledger explorer, courtroom view, verdict report with what-if sliders
Backend	Python with FastAPI ; Server-Sent Events to stream research progress live
Agent orchestration	Custom async orchestrator (asyncio) with a task tree, parallel workers and an event log
LLM layer	LLMs with tool calling and JSON-schema structured output via OpenAI or Groq APIs, chosen on performance, cost and availability
Research & collection	Search APIs (and SearXNG meta-search as fallback), controlled web retrieval, trafilatura for main-text extraction, Web Archive fallback for blocked pages
Evidence storage	PostgreSQL for tasks, claims, quotes, sources, timestamps, verification results and history; pgvector for semantic retrieval
Validation & processing	Embeddings plus MinHash near-duplicate detection for source independence; contradiction and freshness checks in Python
Report generation	Structured Markdown/HTML with optional PDF export , source links and claim-to-source traceability
Dev & deployment	Git and GitHub, Docker , hosting on Render, Railway or AWS

6. Handling edge cases & follow-ups

ResearchOps treats every retrieved result as a **claim to be checked**, not as a fact.

Missing information

A gap is **recorded, not guessed**. The VOI Scheduler creates a **targeted follow-up task** if the gap could change the verdict. If it still cannot be filled, it appears under "**What we could not verify**" and its hypothesis keeps a **wider uncertainty range**. Absence can itself be a finding: for example, no competitor publishing churn figures is reported as such.

Conflicting information

Conflicts are **never averaged away or silently resolved**. Both findings are kept and shown side by side with **date, source type, reliability and independence**. If the conflict affects a **crux**, it becomes a high-value gap and the system searches for a tie-breaking source. Unresolved differences are clearly marked in the report. The Just Walk Out case above is the model example: vendor claims and independent reports would both appear, labelled and weighted differently.

Ambiguous requests

The **Goal Interviewer** identifies what is missing, such as location, time period, target users or success criteria, and asks **open questions** rather than presenting a fixed form. If the user prefers not to answer, ResearchOps proceeds with **explicitly stated assumptions** that stay visible and adjustable as what-if sliders.

A change mid-process

Because research is split into independent tasks and every finding is stored with its source and context, a new requirement **does not restart the investigation**. The Research Director revises the hypotheses, **re-scores existing evidence** without re-fetching, cancels tasks that no longer matter and creates new ones. For example, if a broad market analysis is narrowed to **feasibility for small businesses**, the relevant market and competitor evidence is reused while new tasks cover small-business customers, pricing and operational constraints.

Stale, copied or fabricated sources

Old claims lose weight through **freshness decay** and are labelled with their date. Syndicated articles are merged into **one cluster**. Every claim must carry a **verbatim quote** from a fetched document, so a fabricated URL cannot enter the ledger, and the **verifier** strikes any quote that does not support its claim.

Provider or network failure

Search providers are **health-checked with backoff and fallback**. Fetched pages are **cached with their original retrieval time**, and a completed run can be replayed and clearly marked as cached.

7. What makes this original

The obvious approach is a chatbot that receives a question, searches a few sources, summarises them and attaches links. ResearchOps treats research as a **structured investigation followed by a trial**:

Aspect	Obvious approach	ResearchOps
Output	An essay about the topic	A decision with a probability , cruxes, and must-do / must-not-do actions
User role	Type one prompt	A goal interview that gates the run, plus questions only when they change the outcome
Scope	Agents search freely and drift	Each task has a defined objective and scope set by the Research Director
Evidence	Links attached after writing	A ledger of verbatim quotes built before writing; every sentence cites it
Repetition	Ten copies look like ten sources	Source independence detection : copies count once
Disagreement	Hidden or silently resolved	Shown side by side and argued in the debate
Reasoning	One model, one pass	Adversarial panel with a pre-mortem witness , limited to ledger evidence
Verification	Trust the writer	A separate verifier rules on every claim
Autonomy	Fixed list of sub-questions	Value-of-information : research what would move the decision most
Time	All facts treated as current	Freshness decay per fact type
Exploration	Re-run to test alternatives	What-if sliders from assumptions inferred for this run, recalculated instantly

Each of these targets a **documented weakness** of current research agents and manual research: unsupported or fabricated citations [6][8], false confidence from repeated sources [4], and reports that do not help anyone decide [2][3]. The user sees not only the findings but **how they were reached, how certain they are, and what would change them**.

8. References

- [1] McKinsey Global Institute, "The social economy: Unlocking value and productivity through social technologies", 2012.
- [2] CB Insights, "The Top 20 Reasons Startups Fail", analysis of startup post-mortems, 2014 (updated 2021).
- [3] Quibi shutdown coverage, e.g. The Wall Street Journal and The Verge, October 2020.
- [4] Business Standard, "Amazon's 'just walk out' checkout tech was powered by 1,000 Indian workers", April 2024. https://www.business-standard.com/companies/news/amazon-s-just-walk-out-checkout-tech-was-powered-by-1-000-indian-workers-124040400463_1.html
- [5] The Washington Times, "Amazon's Just Walk Out stores relied on '1,000 people in India'", April 2024, including Amazon's response. <https://www.washingtontimes.com/news/2024/apr/4/amazons-just-walk-out-stores-relied-on-1000-people/>
- [6] CFO Dive, "Deloitte refunds over \$60K for report with AI errors, Australian government", 2025. <https://www.cfodive.com/news/deloitte-refunds-60k-report-ai-errors-australian-government-accounting/803321/> ; AI Incident Database, Incident 1193. <https://incidentdatabase.ai/cite/1193/>
- [7] DeepResearch Bench: A Comprehensive Benchmark for Deep Research Agents, arXiv:2506.11763. <https://arxiv.org/abs/2506.11763>
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- [9] Cited but Not Verified: Parsing and Evaluating Source Attribution, arXiv:2605.06635. <https://arxiv.org/abs/2605.06635>